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Working with Vectors

Every vector is an $\mathbf{i}$ part and a $\mathbf{j}$ part. Sketch it, then Pythagoras for the size and $\tan$ for the direction. Position = start + velocity $\times$ time.

THE TOPIC ON ONE MAP

Working with Vectors

01 — Magnitude, direction, bearings

$|\mathbf{v}|$ angle with $\mathbf{i}$ or $\mathbf{j}$ bearing

02 — Forces as vectors

equilibrium: sum = 0 $\mathbf{F} = m\mathbf{a}$ parallel to a vector

03 — Constant velocity

$\mathbf{r}_0 + \mathbf{v}t$ due east / north of

04 — Two moving objects

$\vec{AB} = \mathbf{b} - \mathbf{a}$ collide closest

THE THREE VECTOR MOVES

size = $\sqrt{a^2 + b^2}$

direction: $\tan \theta = \frac{b}{a}$ from a sketch

parallel to

$p\mathbf{i} + q\mathbf{j}$: $\frac{b}{a} = \frac{q}{p}$

$\mathbf{i}$ east, $\mathbf{j}$ north

bearing: clockwise from north, three figures

01 Magnitude, Direction and Bearings

Always sketch the vector on a small $\mathbf{i}$ - $\mathbf{j}$ cross first. The size comes from Pythagoras, the angle from $\tan$ in the right-angled triangle, and the sketch tells you how to turn that angle into what the question asks (angle with $\mathbf{i}$, angle with $\mathbf{j}$, or a bearing).

WHEN YOU SEE → YOU DO

“exact value of the magnitude”

→ $\sqrt{a^2 + b^2}$ — leave the surd

“angle between $\mathbf{F}$ and $\mathbf{i} + \mathbf{j}$ ”

→ find each direction, then subtract

“ $\mathbf{F}$ acts on a bearing of 120° ”

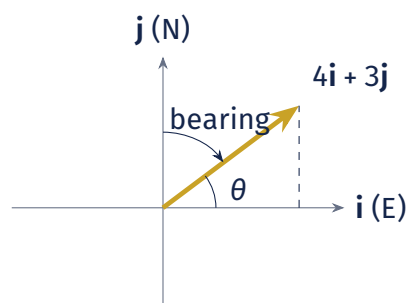
→ $F \sin 120^\circ \mathbf{i} + F \cos 120^\circ \mathbf{j}$

“give the direction as a bearing”

→ sketch; clockwise from north

two forces with an angle between them, no $\mathbf{i}, \mathbf{j}$

→ triangle of forces: cosine rule, sine rule



$$\text{size} = 5, \tan \theta = \frac{3}{4}, \theta = 36.9^\circ$$

$$\text{bearing} = 90^\circ - 36.9^\circ = 053^\circ$$

WORKED MODEL • JUNE 2024

P: 10 N due west. **Q:** 8 N on a bearing of 330° . Find $|\mathbf{P} + \mathbf{Q}|$.

1 $\mathbf{Q} = 8 \sin 330^\circ \mathbf{i} + 8 \cos 330^\circ \mathbf{j} = -4\mathbf{i} + 4\sqrt{3}\mathbf{j}$

bearing to $\mathbf{i}, \mathbf{j}$

2 $\mathbf{F} = -14\mathbf{i} + 4\sqrt{3}\mathbf{j}$

add

ANSWER $|\mathbf{F}| = \sqrt{196 + 48} = 2\sqrt{61} \approx 15.6 \text{ N}$

IDEA 01

Practise: Magnitude, Direction and Bearings

My questions first, then real past papers.

Q1 **DR ESLAM** Magnitude and bearing

M1 • TOPIC 2 3 marks

$\mathbf{i}$ is east, $\mathbf{j}$ is north. Find the magnitude of $\mathbf{F} = (-4\mathbf{i} + 3\mathbf{j})$ N and its bearing. Sketch it first: west 4, north 3.

Q2 **DR ESLAM** From a bearing to $\mathbf{i}, \mathbf{j}$

M1 • TOPIC 2 2 marks

A force of 10 N acts on a bearing of 060° . Write it in the form $a\mathbf{i} + b\mathbf{j}$.

Q3 **DR ESLAM** Angle between two vectors

M1 • TOPIC 2 3 marks

Find the angle between $3\mathbf{i} + 4\mathbf{j}$ and $\mathbf{i} + \mathbf{j}$.

PAST PAPERS

Q4 PAST PAPER Pearson IAL WME01 • October 2019 • Q7

M1 • TOPIC 2 7 marks

7. Two forces, **F** and **G**, act on a particle. The force **F** has magnitude 4 N and acts in a direction with a bearing of 120° and the force **G** has magnitude 6 N and acts due north.

Given that $\mathbf{P} = 2\mathbf{F} + \mathbf{G}$, find

- (i) the magnitude of **P**
- (ii) the direction of **P**, giving your answer as a bearing to the nearest degree.

(7)

Q5 **PAST PAPER** Pearson IAL WME01 • January 2020 • Q6**M1 • TOPIC 2** **11 marks**

6. A force $\mathbf{F}$ is given by $\mathbf{F} = (10\mathbf{i} + \mathbf{j})\text{N}$.

(a) Find the exact value of the magnitude of $\mathbf{F}$.

(2)

(b) Find, in degrees, the size of the angle between the direction of $\mathbf{F}$ and the direction of the vector $(\mathbf{i} + \mathbf{j})$.

(4)

The resultant of the force $\mathbf{F}$ and the force $(-15\mathbf{i} + a\mathbf{j})\text{N}$, where a is a constant, is parallel to, but in the opposite direction to, the vector $(2\mathbf{i} - 3\mathbf{j})$.

(c) Find the value of a .

(5)

Q6 **PAST PAPER** Pearson IAL WME01 • January 2021 • Q5**M1 • TOPIC 2** **7 marks**

5. A particle is acted upon by two forces **F** and **G**. The force **F** has magnitude 8 N and acts in a direction with a bearing of 240° . The force **G** has magnitude 10 N and acts due South.

Given that $\mathbf{R} = \mathbf{F} + \mathbf{G}$, find

- (i) the magnitude of **R**,
- (ii) the direction of **R**, giving your answer as a bearing to the nearest degree.

(7)

Q7**PAST PAPER**

Pearson IAL WME01 • January 2024 • Q4

M1 • TOPIC 2 8 marks

4.

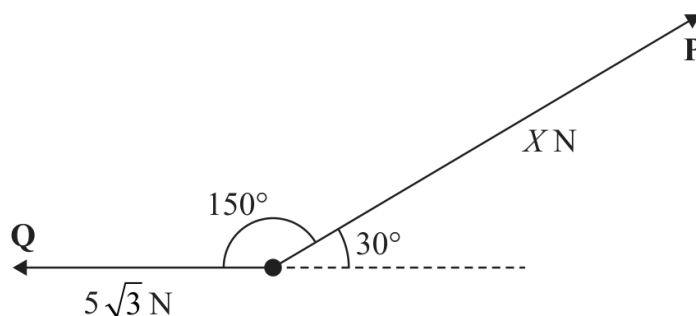
**Figure 3**

Figure 3 shows two horizontal forces **P** and **Q** acting on a particle.

The angle between the direction of **P** and the direction of **Q** is 150°

Force **P** has magnitude X newtons.

Force **Q** has magnitude $5\sqrt{3}$ N.

The resultant of **P** and **Q** has magnitude $\sqrt{129}$ N.

Find

- (i) the value of X .
- (ii) the angle between **Q** and the resultant, giving your answer to the nearest degree.

(8)

Q8 **PAST PAPER** Pearson IAL WME01 · June 2024 · Q2**M1 · TOPIC 2** **4 marks**

2. Two forces, **P** and **Q**, act on a particle.

- **P** has magnitude 10 N and acts due west
- **Q** has magnitude 8 N and acts on a bearing of 330°

Given that $\mathbf{F} = \mathbf{P} + \mathbf{Q}$, find the magnitude of **F**.

(4)

Q9**PAST PAPER**

Pearson IAL WME01 · June 2026 · Q6

M1 · TOPIC 2**9 marks**

6. The resultant $\mathbf{R}$ of two forces $\mathbf{F}_1 \text{ N}$ and $\mathbf{F}_2 \text{ N}$ is $(-5\mathbf{i} + 6\mathbf{j}) \text{ N}$.

(a) Find the magnitude of $\mathbf{R}$.

(2)

(b) Find, in degrees, the angle between the direction of $\mathbf{R}$ and the direction of the vector $(2\mathbf{i} + \mathbf{j})$. Show your method clearly.

(3)

Given that $\mathbf{F}_1$ is parallel to the vector $(-\mathbf{i} + 2\mathbf{j})$ and that $\mathbf{F}_2$ is parallel to the vector $(2\mathbf{i} - 3\mathbf{j})$,

(c) find $\mathbf{F}_1$ and $\mathbf{F}_2$

(4)

(Total for Question 6 is 9 marks)

02 Forces as Vectors: Equilibrium and $F = ma$

Add all the forces as vectors. In equilibrium the sum is $\mathbf{0}$ (the $\mathbf{i}$ total is 0 and the $\mathbf{j}$ total is 0). Otherwise the sum is ma . “In the direction of $p\mathbf{i} + q\mathbf{j}$ ” gives a ratio equation.

WHEN YOU SEE → YOU DO

“rests in equilibrium under three forces”

$$\rightarrow \mathbf{F}_1 + \mathbf{F}_2 + \mathbf{F}_3 = \mathbf{0}$$

“the acceleration has magnitude 3.25”

$$\rightarrow |\mathbf{R}| = m \times 3.25$$

“the resultant is parallel to $(2\mathbf{i} - 3\mathbf{j})$ ”

$$\rightarrow \mathbf{R} = \lambda(2\mathbf{i} - 3\mathbf{j}) \text{ — compare parts}$$

“in the opposite direction to”

$$\rightarrow \lambda < 0$$

“moves in the direction of $(\mathbf{i} - 2\mathbf{j})$ ” from rest

$$\rightarrow \text{the resultant force has that direction}$$

Add the forces

Sum = 0 or ma

Compare $\mathbf{i}$ and $\mathbf{j}$ parts

Solve

WORKED MODEL • JANUARY 2026

$\mathbf{F}_1 = 2p\mathbf{i} - 3p\mathbf{j}$, $\mathbf{F}_2 = q\mathbf{i} + \mathbf{j}$, $\mathbf{F}_3 = 3\mathbf{i} - q\mathbf{j}$. The resultant is parallel to $4\mathbf{i} - 5\mathbf{j}$. Show $2p - q = 19$.

1 $\mathbf{R} = (2p + q + 3)\mathbf{i} + (1 - 3p - q)\mathbf{j}$

add

2 $-5(2p + q + 3) = 4(1 - 3p - q)$

parts in the ratio 4 : -5

ANSWER $2p - q = 19$

× **TRAP** Writing “parallel” as “equal”. Parallel means a multiple: compare the ratio of the parts.

IDEA 02

Practise: Forces as Vectors: Equilibrium and $F = ma$

My questions first, then real past papers.

Q10 DR ESLAM Three forces in equilibrium

M1 · TOPIC 2 2 marks

$(2\mathbf{i} - \mathbf{j})$, $(p\mathbf{i} + 3\mathbf{j})$ and $(-5\mathbf{i} + q\mathbf{j})$ N are in equilibrium. Find p and q .

Q11 DR ESLAM Acceleration from forces

M1 · TOPIC 2 3 marks

A 2 kg particle is acted on by $(3\mathbf{i} + 2\mathbf{j})$ N and $(\mathbf{i} - 6\mathbf{j})$ N. Find its acceleration and the magnitude.

Q12 DR ESLAM Parallel to a vector

M1 · TOPIC 2 3 marks

The resultant of $(p\mathbf{i} + 2\mathbf{j})$ and $(3\mathbf{i} + p\mathbf{j})$ is parallel to $\mathbf{i} + 2\mathbf{j}$. Find p .
Parallel to $\mathbf{i} + 2\mathbf{j}$: the $\mathbf{j}$ part is twice the $\mathbf{i}$ part.

PAST PAPERS

Q13 PAST PAPER Pearson IAL WME01 • June 2021 • Q3

M1 • TOPIC 2 9 marks

3. [In this question $\mathbf{i}$ and $\mathbf{j}$ are perpendicular horizontal unit vectors.]

Three forces, $\mathbf{F}_1$, $\mathbf{F}_2$ and $\mathbf{F}_3$, are given by

$$\mathbf{F}_1 = (5\mathbf{i} + 2\mathbf{j})\text{N} \quad \mathbf{F}_2 = (-3\mathbf{i} + \mathbf{j})\text{N} \quad \mathbf{F}_3 = (a\mathbf{i} + b\mathbf{j})\text{N}$$

where a and b are constants.

The forces $\mathbf{F}_1$, $\mathbf{F}_2$ and $\mathbf{F}_3$ act on a particle P of mass 4 kg.

Given that P rests in equilibrium on a smooth horizontal surface under the action of these three forces,

- (a) find the size of the angle between the direction of $\mathbf{F}_3$ and the direction of $-\mathbf{j}$. (4)

The force $\mathbf{F}_3$ is now removed and replaced by the force $\mathbf{F}_4$ given by $\mathbf{F}_4 = \lambda(\mathbf{i} + 3\mathbf{j})\text{N}$, where λ is a positive constant.

When the three forces $\mathbf{F}_1$, $\mathbf{F}_2$ and $\mathbf{F}_4$ act on P , the acceleration of P has magnitude 3.25 ms^{-2}

- (b) Find the value of λ . (5)

Q14 PAST PAPER Pearson IAL WME01 • January 2022 • Q6

M1 • TOPIC 2 12 marks

6. [In this question $\mathbf{i}$ and $\mathbf{j}$ are horizontal unit vectors.]

A particle P of mass 2 kg moves under the action of two forces, $(p\mathbf{i} + q\mathbf{j})\text{ N}$ and $(2q\mathbf{i} + p\mathbf{j})\text{ N}$, where p and q are constants.

Given that the acceleration of P is $(\mathbf{i} - \mathbf{j})\text{ m s}^{-2}$

(a) find the value of p and the value of q .

(5)

(b) Find the size of the angle between the direction of the acceleration and the vector $\mathbf{j}$.

(2)

At time $t = 0$, the velocity of P is $(3\mathbf{i} - 4\mathbf{j})\text{ m s}^{-1}$

At $t = T$ seconds, P is moving in the direction of the vector $(11\mathbf{i} - 13\mathbf{j})$.

(c) Find the value of T .

(5)

Q15**PAST PAPER**

Pearson IAL WME01 · October 2022 · Q6

M1 · TOPIC 2 9 marks

6. [In this question, $\mathbf{i}$ and $\mathbf{j}$ are horizontal unit vectors.]

A particle A of mass 0.5 kg is at rest on a smooth horizontal plane.

At time $t = 0$, two forces, $\mathbf{F}_1 = (-3\mathbf{i} + 2\mathbf{j})\text{ N}$ and $\mathbf{F}_2 = (p\mathbf{i} + q\mathbf{j})\text{ N}$, where p and q are constants, are applied to A .

Given that A moves in the direction of the vector $(\mathbf{i} - 2\mathbf{j})$,

(a) show that $2p + q - 4 = 0$ (4)

Given that $p = 5$

(b) find the speed of A at time $t = 4$ seconds. (5)

Q16 PAST PAPER Pearson IAL WME01 · January 2023 · Q3

M1 · TOPIC 2 10 marks

3. A particle P is moving with constant acceleration $(-4\mathbf{i} + \mathbf{j})\text{ms}^{-2}$

At time $t = 0$, P has velocity $(14\mathbf{i} - 5\mathbf{j})\text{ms}^{-1}$

- (a) Find the speed of P at time $t = 2$ seconds.

(3)

- (b) Find the size of the angle between the direction of $\mathbf{i}$ and the direction of motion of P at time $t = 2$ seconds.

(3)

At time $t = T$ seconds, P is moving in the direction of vector $(2\mathbf{i} - 3\mathbf{j})$

- (c) Find the value of T

(4)

Q17 PAST PAPER Pearson IAL WME01 • June 2023 • Q2

M1 • TOPIC 2 10 marks

[In this question $\mathbf{i}$ and $\mathbf{j}$ are horizontal perpendicular unit vectors.]

2. A particle P rests in equilibrium on a smooth horizontal plane.

A system of **three** forces, $\mathbf{F}_1 \text{ N}$, $\mathbf{F}_2 \text{ N}$ and $\mathbf{F}_3 \text{ N}$ where

$$\mathbf{F}_1 = (3c\mathbf{i} + 4c\mathbf{j})$$

$$\mathbf{F}_2 = (-14\mathbf{i} + 7\mathbf{j})$$

is applied to P .

Given that P remains in equilibrium,

- (a) find $\mathbf{F}_3$ in terms of c , $\mathbf{i}$ and $\mathbf{j}$.

(2)

The force $\mathbf{F}_3$ is **removed** from the system.

Given that $c = 2$

- (b) find the size of the angle between the direction of $\mathbf{i}$ and the direction of the resultant force acting on P .

(4)

The mass of P is $m \text{ kg}$.

Given that the magnitude of the acceleration of P is 8.5 m s^{-2}

- (c) find the value of m .

(4)

Q18 **PAST PAPER** Pearson IAL WME01 · January 2025 · Q1**M1 · TOPIC 2** **8 marks**

1. A particle of mass 2.5 kg moves on a smooth horizontal plane under the action of three horizontal forces, $\mathbf{F}_1$, $\mathbf{F}_2$ and $\mathbf{F}_3$, where

$$\mathbf{F}_1 = (6\mathbf{i} + 8\mathbf{j})\text{N}$$

$$\mathbf{F}_2 = (-16\mathbf{i} + 2\mathbf{j})\text{N}$$

$$\mathbf{F}_3 = (-2\mathbf{i} + 8\mathbf{j})\text{N}$$

- (a) Find the magnitude of the acceleration of the particle.

(4)

A fourth force, $\mathbf{F}_4 = (p\mathbf{i} + p\mathbf{j})\text{N}$, where p is a constant, is added.

The resultant of the four forces acts in the direction of the vector $(7\mathbf{i} + 2\mathbf{j})$.

- (b) Find the value of p .

(4)

Q19**PAST PAPER**

Pearson IAL WME01 • October 2025 • Q3

M1 • TOPIC 2 14 marks

3. [In this question $\mathbf{i}$ and $\mathbf{j}$ are horizontal perpendicular unit vectors.]

A particle P of mass 2 kg moves on a smooth horizontal surface under the action of two forces $\mathbf{F}_1$ and $\mathbf{F}_2$, where $\mathbf{F}_1 = (-2\mathbf{i} + 3\mathbf{j})\text{ N}$ and $\mathbf{F}_2 = (4\mathbf{i} + 2\mathbf{j})\text{ N}$.

(a) Find the acceleration of P .

(3)

At time $t = 0$, the velocity of P is $(3\mathbf{i} - 4\mathbf{j})\text{ m s}^{-1}$

(b) Find the speed of P when $t = 3$ seconds.

(4)

An additional force, $\mathbf{F}_3 = (b\mathbf{i} + c\mathbf{j})\text{ N}$, is applied to P .

The resultant of $\mathbf{F}_1$, $\mathbf{F}_2$ and $\mathbf{F}_3$ is equal to $\lambda(\mathbf{i} + \mathbf{j})\text{ N}$, where λ is a constant.

(c) Show that $b - c = 3$

(3)

The resultant of $\mathbf{F}_1$, $\mathbf{F}_2$ and $\mathbf{F}_3$ has magnitude $10\sqrt{2}\text{ N}$.

(d) Find the two possible $\mathbf{F}_3$ forces.

(4)

Q20 **PAST PAPER** Pearson IAL WME01 • January 2026 • Q3**M1 • TOPIC 2** **8 marks**

3. A particle moves on a smooth horizontal plane under the action of three horizontal forces, $\mathbf{F}_1$, $\mathbf{F}_2$ and $\mathbf{F}_3$, where

$$\mathbf{F}_1 = (2p\mathbf{i} - 3p\mathbf{j})\text{N}$$

$$\mathbf{F}_2 = (q\mathbf{i} + \mathbf{j})\text{N}$$

$$\mathbf{F}_3 = (3\mathbf{i} - q\mathbf{j})\text{N}$$

where p and q are constants.

The resultant force acts in the direction of the vector $(4\mathbf{i} - 5\mathbf{j})$

- (a) Show that

$$2p - q = 19 \quad (4)$$

The mass of the particle is 0.5 kg.

Given that $p = 7$

- (b) find the acceleration of the particle. (4)

03 Constant Velocity: $\mathbf{r} = \mathbf{r}_0 + \mathbf{v}t$

Position at time t = where it started + velocity $\times t$. If you know two positions, the velocity is (change of position) $\div$ (time between them). “Due east of O ” means the $\mathbf{j}$ part is 0.

WHEN YOU SEE $\rightarrow$ YOU DO

“at 7 am ... at 7.36 am”

$\rightarrow$ time in hours: 0.6 h

“find the velocity” from two positions

$$\rightarrow \mathbf{v} = \frac{\mathbf{r}_2 - \mathbf{r}_1}{t_2 - t_1}$$

“due east / due north of O ”

$\rightarrow$ $\mathbf{j}$ part = 0 / $\mathbf{i}$ part = 0

“north-east of O ”

$\rightarrow$ $\mathbf{i}$ part = $\mathbf{j}$ part (both positive)

“distance from O is 9 km or less”

$\rightarrow |\mathbf{r}|^2 \leq 81$ — a quadratic in t

“ OP perpendicular to the direction of motion”

$\rightarrow \mathbf{r} \cdot \mathbf{v} = 0$, or the closest-point method

WORKED MODEL • JUNE 2025

$\mathbf{r} = -5\mathbf{i} + 16\mathbf{j}$ at $t = 2$ and $10\mathbf{i} + 4\mathbf{j}$ at $t = 5$. Find $\mathbf{v}$ and $\mathbf{r}$ at time t .

1 $\mathbf{v} = \frac{15\mathbf{i} - 12\mathbf{j}}{3} = 5\mathbf{i} - 4\mathbf{j}$

change $\div$ time

2 $\mathbf{r} = (-5\mathbf{i} + 16\mathbf{j}) + (t - 2)(5\mathbf{i} - 4\mathbf{j})$

start at $t = 2$

ANSWER $\mathbf{r} = (5t - 15)\mathbf{i} + (24 - 4t)\mathbf{j}$

IDEA 03

Practise: Constant Velocity: $\mathbf{r} = \mathbf{r}_0 + \mathbf{v}t$

My questions first, then real past papers.

Q21 DR ESLAM Where and when

M1 • TOPIC 2 3 marks

A boat starts at $(2\mathbf{i} + 5\mathbf{j})$ km with velocity $(3\mathbf{i} - \mathbf{j})$ km h⁻¹. Find its position after t hours and when it is due east of O .

Q22 DR ESLAM Velocity from two positions

M1 • TOPIC 2 2 marks

A particle passes $(\mathbf{i} + 2\mathbf{j})$ m at $t = 0$ and $(7\mathbf{i} - 6\mathbf{j})$ m at $t = 2$. Find its velocity and speed.

PAST PAPERS

Q23 PAST PAPER Pearson IAL WME01 • October 2021 • Q4

M1 • TOPIC 2 8 marks

4. The position vector, $\mathbf{r}$ metres, of a particle P at time t seconds, relative to a fixed origin O , is given by

$$\mathbf{r} = (t - 3)\mathbf{i} + (1 - 2t)\mathbf{j}$$

- (a) Find, to the nearest degree, the size of the angle between $\mathbf{r}$ and the vector $\mathbf{j}$, when $t = 2$
(3)

- (b) Find the values of t for which the distance of P from O is 2.5 m.
(5)

Q24 PAST PAPER Pearson IAL WME01 • October 2021 • Q8**M1 • TOPIC 2 12 marks**

8. [In this question $\mathbf{i}$ and $\mathbf{j}$ are horizontal unit vectors directed due east and due north respectively and position vectors are given relative to a fixed origin.]

At 7 am a ship leaves a port and moves with constant velocity. The position vector of the port is $(-2\mathbf{i} + 9\mathbf{j})$ km.

At 7.36 am the ship is at the point with position vector $(4\mathbf{i} + 6\mathbf{j})$ km.

- (a) Show that the velocity of the ship is $(10\mathbf{i} - 5\mathbf{j}) \text{ km h}^{-1}$ (2)

- (b) Find the position vector of the ship t hours after leaving port. (2)

At 8.48 am a passenger on the ship notices that a lighthouse is due east of the ship.

At 9 am the same passenger notices that the lighthouse is now north east of the ship.

- (c) Find the position vector of the lighthouse. (4)

- (d) Find the position vector of the ship when it is due south of the lighthouse. (4)

(Total 12 marks)

Q25**PAST PAPER**

Pearson IAL WME01 · October 2024 · Q3

M1 · TOPIC 2**11 marks**

3. [In this question $\mathbf{i}$ and $\mathbf{j}$ are horizontal unit vectors and position vectors are given relative to a fixed origin.]

A ship A is moving with constant velocity.

At 1 pm, the position vector of A is $(25\mathbf{i} + 10\mathbf{j})$ km.

At 3 pm, the position vector of A is $(55\mathbf{i} + 34\mathbf{j})$ km.

At time t hours after 1 pm, the position vector of A is $\mathbf{r}_A$ km.

- (a) Show that $\mathbf{r}_A = (25 + 15t)\mathbf{i} + (10 + 12t)\mathbf{j}$ (4)

The speed of A is $V \text{ ms}^{-1}$

- (b) Find the value of V . (2)

A ship B is moving with constant velocity $(20\mathbf{i} - 6\mathbf{j}) \text{ km h}^{-1}$

At 1 pm, the position vector of B is $(35\mathbf{i} + 51\mathbf{j})$ km.

At 2:30 pm, B passes through the point P .

- (c) Show that A also passes through P . (5)

Q26**PAST PAPER**

Pearson IAL WME01 · January 2025 · Q5

M1 · TOPIC 2 7 marks

5. [In this question $\mathbf{i}$ and $\mathbf{j}$ are horizontal perpendicular unit vectors and position vectors are given relative to a fixed origin.]

In a game, a ball B is rolled across a horizontal surface towards a fixed target.
The ball is modelled as a particle moving with constant velocity.

At time $t = 1$ s, the position vector of B is $(-2\mathbf{i} + 5\mathbf{j})$ m.

At time $t = 9$ s, the position vector of B is $(10\mathbf{i} - 3\mathbf{j})$ m.

- (a) Find the velocity of the ball.

(3)

The position vector of the target is $(13\mathbf{i} - 5\mathbf{j})$ m.

- (b) Use the model to find the distance of B from the target at time $t = 7$ s.

(4)

Q27**PAST PAPER**

Pearson IAL WME01 · June 2025 · Q2

M1 · TOPIC 2 7 marks

2. [In this question $\mathbf{i}$ and $\mathbf{j}$ are horizontal unit vectors directed due east and due north respectively. Position vectors are given relative to a fixed origin.]

Two particles, A and B , are moving on a smooth horizontal surface.
Each particle is moving with constant velocity.

At time t seconds, the position vector of A is given by $\mathbf{r}$ metres.

- When $t = 2$, $\mathbf{r} = (-5\mathbf{i} + 16\mathbf{j})$
- When $t = 5$, $\mathbf{r} = (10\mathbf{i} + 4\mathbf{j})$

(a) Find, in terms of $\mathbf{i}$ and $\mathbf{j}$, the velocity of A .

(2)

(b) Find an expression for $\mathbf{r}$ at time t seconds.

Give your answer in the form $p\mathbf{i} + q\mathbf{j}$, where p and q are functions of t

(2)

At time t seconds, the position vector of B is given by $\mathbf{s}$ metres where

$$\mathbf{s} = -5\mathbf{i} + 7\mathbf{j} + t(2\mathbf{i} - 3\mathbf{j})$$

(c) Find, to the nearest degree, the bearing of B from A when $t = 5$

(3)

Q28**PAST PAPER**

Pearson IAL WME01 · June 2026 · Q7

M1 · TOPIC 2 15 marks

7. [In this question $\mathbf{i}$ and $\mathbf{j}$ are horizontal unit vectors directed due east and due north respectively and position vectors are given relative to a fixed origin O .]

A boat, which is modelled as a particle, is moving with constant velocity $(-2\mathbf{i} + \mathbf{j}) \text{ km h}^{-1}$

At time $t = 1.5$ hours after noon, the boat passes through the point with position vector $(11\mathbf{i} + 3.5\mathbf{j}) \text{ km}$.

- (a) Show that, at time t hours after noon, the position vector of the boat is

$$[(14 - 2t)\mathbf{i} + (2 + t)\mathbf{j}] \text{ km} \quad (3)$$

The boat is visible from O when it is 9 km or less from O .

- (b) Find the length of time for which the boat is visible from O , showing your method clearly. (4)

- (c) Find the value of t when the boat is northwest of O . (2)

At the instant when the boat is at the point P , OP is perpendicular to the direction in which the boat is moving.

- (d) Show that $OP = \frac{18\sqrt{5}}{5} \text{ km}$

(Solutions relying on calculator technology are not acceptable.) (6)

(Total for Question 7 is 15 marks)

04 Two Moving Objects: Meet, Closest, Bearing

Write both position vectors in t . The vector from A to B is $\vec{AB} = \mathbf{b} - \mathbf{a}$. They collide only if *both* parts are zero at the *same* time. Their distance is $|\vec{AB}|$; the closest approach is the minimum of $|\vec{AB}|^2$ (a quadratic in t — complete the square).

WHEN YOU SEE → YOU DO

“show that $\vec{AB} = \dots$ ”

→ $\mathbf{b} - \mathbf{a}$, collect $\mathbf{i}$ and $\mathbf{j}$

“verify that they would collide at 13:30”

→ same position at $t = 1.5$

“show that they never collide”

→ the two parts are not zero together

“shortest distance between the ships”

→ minimise $|\vec{AB}|^2$: complete the square

“the bearing of B from A ”

→ sketch $\vec{AB}$, measure from north at A

“ B is north-east of A ”

→ $\vec{AB}$ has equal positive parts

WORKED MODEL • JANUARY 2022

$\vec{AB} = (12t - 10)\mathbf{i} + (24t - 10)\mathbf{j}$. Find the shortest distance.

$$1 \quad |\vec{AB}|^2 = (12t - 10)^2 + (24t - 10)^2 = 720t^2 - 720t + 200$$

square and add

$$2 \quad = 720\left(t - \frac{1}{2}\right)^2 + 20$$

complete the square

ANSWER least $|AB| = \sqrt{20} = 2\sqrt{5} \approx 4.5 \text{ km}$ at $t = \frac{1}{2}$

× **TRAP** Saying they collide because the $\mathbf{i}$ parts are equal at one time. Both parts must match at that same time.

IDEA 04

Practise: Two Moving Objects: Meet, Closest, Bearing

My questions first, then real past papers.

Q29 DR ESLAM Collide or not? Closest?

M1 • TOPIC 2 5 marks

A starts at O with velocity $(4\mathbf{i} + 2\mathbf{j})$; B starts at $10\mathbf{i}$ with velocity $(2\mathbf{i} + 4\mathbf{j})$ (km, h). Find $\vec{AB}$, decide whether they collide, and find their least distance apart.

$\vec{AB} = \mathbf{b} - \mathbf{a}$. Collide: both parts zero at the SAME t .

PAST PAPERS

Q30 PAST PAPER Pearson IAL WME01 · October 2019 · Q8

M1 · TOPIC 2 14 marks

8. [In this question, the horizontal unit vectors $\mathbf{i}$ and $\mathbf{j}$ are directed due east and due north respectively and position vectors are given relative to a fixed origin O .]

Two speedboats, A and B , are each moving with constant velocity. The velocity of A is 20 km h^{-1} due west and the velocity of B is 40 km h^{-1} on a bearing of 150° . The boats are modelled as particles.

At noon, the position vector of A is $60\mathbf{i} \text{ km}$ and B is at the origin O . At time t hours after noon, the position vector of A is $\mathbf{r} \text{ km}$ and the position vector of B is $\mathbf{s} \text{ km}$.

- (a) Find the velocity of B in the form $(p\mathbf{i} + q\mathbf{j}) \text{ km h}^{-1}$ (3)
- (b) Find expressions for $\mathbf{r}$ and $\mathbf{s}$ in terms of t , $\mathbf{i}$ and $\mathbf{j}$. (3)
- (c) Find the time, to the nearest minute, at which the distance between the boats is the same as it was at noon. (8)

Q31 PAST PAPER Pearson IAL WME01 • January 2021 • Q6

M1 • TOPIC 2 12 marks

6. Two girls, Agatha and Brionie, are roller skating inside a large empty building. The girls are modelled as particles.

At time $t = 0$, Agatha is at the point with position vector $(11\mathbf{i} + 11\mathbf{j})\text{m}$ and Brionie is at the point with position vector $(7\mathbf{i} + 16\mathbf{j})\text{m}$. The position vectors are given relative to the door, O , and $\mathbf{i}$ and $\mathbf{j}$ are horizontal perpendicular unit vectors.

Agatha skates with constant velocity $(3\mathbf{i} - \mathbf{j})\text{m s}^{-1}$

Brionie skates with constant velocity $(4\mathbf{i} - 2\mathbf{j})\text{m s}^{-1}$

- (a) Find the position vector of Agatha at time t seconds.

(2)

At time $t = 6$ seconds, Agatha passes through the point P .

- (b) Show that Brionie also passes through P and find the value of t when this occurs.

(4)

At time t seconds, Agatha is at the point A and Brionie is at the point B .

- (c) Show that $\overrightarrow{AB} = [(t - 4)\mathbf{i} + (5 - t)\mathbf{j}]\text{m}$

(2)

- (d) Find the distance between the two girls when they are closest together.

(4)

Q32**PAST PAPER**

Pearson IAL WME01 · January 2022 · Q8

M1 · TOPIC 2 14 marks

8. [In this question $\mathbf{i}$ and $\mathbf{j}$ are horizontal unit vectors directed due east and due north respectively and position vectors are given relative to a fixed origin.]

A ship A moves with constant velocity $(3\mathbf{i} - 10\mathbf{j}) \text{ km h}^{-1}$

At time t hours, the position vector of A is $\mathbf{r}$ km.

At time $t = 0$, A is at the point with position vector $(13\mathbf{i} + 5\mathbf{j}) \text{ km}$.

- (a) Find $\mathbf{r}$ in terms of t .

(2)

Another ship B moves with constant velocity $(15\mathbf{i} + 14\mathbf{j}) \text{ km h}^{-1}$

At time $t = 0$, B is at the point with position vector $(3\mathbf{i} - 5\mathbf{j}) \text{ km}$.

- (b) Show that, at time t hours,

$$\overrightarrow{AB} = [(12t - 10)\mathbf{i} + (24t - 10)\mathbf{j}] \text{ km}$$

(4)

Given that the two ships do not change course,

- (c) find the shortest distance between the two ships,

(6)

- (d) find the bearing of ship B from ship A when the ships are closest.

(2)**(Total 14 marks)**

Q33**PAST PAPER**

Pearson IAL WME01 • June 2022 • Q8

M1 • TOPIC 2 17 marks

8. [In this question, $\mathbf{i}$ and $\mathbf{j}$ are horizontal unit vectors directed due east and due north respectively and position vectors are given relative to a fixed origin O .]

Two boats, P and Q , are moving with constant velocities.

The velocity of P is $15\mathbf{i} \text{ m s}^{-1}$ and the velocity of Q is $(20\mathbf{i} - 20\mathbf{j}) \text{ m s}^{-1}$

- (a) Find the direction in which Q is travelling, giving your answer as a bearing.

(2)

The boats are modelled as particles.

At time $t = 0$, P is at the origin O and Q is at the point with position vector $200\mathbf{j} \text{ m}$.

At time t seconds, the position vector of P is $\mathbf{p} \text{ m}$ and the position vector of Q is $\mathbf{q} \text{ m}$.

- (b) Show that

$$\overrightarrow{PQ} = [5t\mathbf{i} + (200 - 20t)\mathbf{j}] \text{ m} \quad (5)$$

- (c) Find the bearing of P from Q when $t = 10$

(2)

- (d) Find the distance between P and Q when Q is north east of P

(5)

- (e) Find the times when P and Q are 200 m apart.

(3)**(Total 17 marks)**

Q34 PAST PAPER Pearson IAL WME01 · October 2022 · Q8

M1 · TOPIC 2 16 marks

8. [In this question, $\mathbf{i}$ and $\mathbf{j}$ are horizontal unit vectors directed due east and due north respectively and position vectors are given relative to a fixed origin O .]

Two ships, A and B , are moving with constant velocities.

The velocity of A is $(3\mathbf{i} + 12\mathbf{j})\text{ km h}^{-1}$ and the velocity of B is $(p\mathbf{i} + q\mathbf{j})\text{ km h}^{-1}$

- (a) Find the speed of A .

(2)

The ships are modelled as particles.

At 12 noon, A is at the point with position vector $(-9\mathbf{i} + 6\mathbf{j})\text{ km}$ and B is at the point with position vector $(16\mathbf{i} + 6\mathbf{j})\text{ km}$.

At time t hours after 12 noon,

$$\overrightarrow{AB} = [(25 - 12t)\mathbf{i} - 9t\mathbf{j}]\text{ km}$$

- (b) Find the value of p and the value of q .

(7)

- (c) Find the bearing of A from B when the ships are 15 km apart, giving your answer to the nearest degree.

(7)

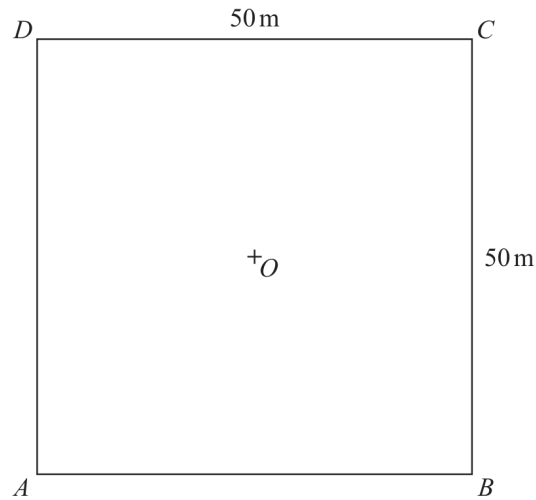
(Total 16 marks)

Q35**PAST PAPER**

Pearson IAL WME01 • June 2023 • Q8

M1 • TOPIC 2 9 marks

8.

**Figure 5**

A square floor space $ABCD$, with centre O , is modelled as a flat horizontal surface measuring 50 m by 50 m, as shown in Figure 5.

The horizontal unit vectors $\mathbf{i}$ and $\mathbf{j}$ are in the direction of $\overrightarrow{AB}$ and $\overrightarrow{AD}$ respectively.

All position vectors are given relative to O .

A small robot R is programmed to travel across the floor at a constant velocity.

- At time $t = 0$, R is at the point with position vector $(-2\mathbf{i} + \mathbf{j})$ m
- At time $t = 11$ s, R is at the point with position vector $(9\mathbf{i} + 23\mathbf{j})$ m
- At time t seconds, the position vector of R is $\mathbf{r}$ metres

(a) Find, in terms of t , $\mathbf{i}$ and $\mathbf{j}$, an expression for $\mathbf{r}$

(3)

A second robot S is at the point C .

- At time $t = 0$, S leaves C and moves with constant velocity $(-\mathbf{i} - \mathbf{j})$ m s⁻¹
- At time t seconds, the position vector of S is $\mathbf{s}$ metres

(b) Write down, in terms of t , $\mathbf{i}$ and $\mathbf{j}$, an expression for $\mathbf{s}$

(1)

(c) Show that

$$\overrightarrow{SR} = [(2t - 27)\mathbf{i} + (3t - 24)\mathbf{j}] \text{ m}$$

(2)

(d) Find the time when the distance between R and S is a minimum.

(3)

Q36 PAST PAPER Pearson IAL WME01 · October 2023 · Q6

M1 · TOPIC 2 15 marks

6. [In this question $\mathbf{i}$ and $\mathbf{j}$ are horizontal unit vectors directed due east and due north respectively and position vectors are given relative to a fixed origin O .]

At 12:00, a ship P sets sail from a harbour with position vector $(15\mathbf{i} + 36\mathbf{j})$ km.

At 12:30, P is at the point with position vector $(20\mathbf{i} + 34\mathbf{j})$ km.

Given that P moves with constant velocity,

- (a) show that the velocity of P is $(10\mathbf{i} - 4\mathbf{j}) \text{ km h}^{-1}$ (2)

At time t hours after 12:00, the position vector of P is $\mathbf{p}$ km.

- (b) Find an expression for $\mathbf{p}$ in terms of $\mathbf{i}$, $\mathbf{j}$ and t . (2)

A second ship Q is also travelling at a constant velocity.

At time t hours after 12:00, the position vector of Q is given by $\mathbf{q}$ km, where

$$\mathbf{q} = (42 - 8t)\mathbf{i} + (9 + 14t)\mathbf{j}$$

Ships P and Q are modelled as particles.

If both ships maintained their course,

- (c) (i) verify that they would collide at 13:30
(ii) find the position vector of the point at which the collision would occur. (4)

At 12:30 Q changes speed and direction to avoid the collision.

Ship Q now travels due north with a constant speed of 15 km h^{-1}

Ship P maintains the same constant velocity throughout.

- (d) Find the exact distance between P and Q at 14:30 (7)

Q37 PAST PAPER Pearson IAL WME01 • January 2024 • Q7

M1 • TOPIC 2 11 marks

7. [In this question $\mathbf{i}$ and $\mathbf{j}$ are horizontal unit vectors directed due east and due north respectively and position vectors are given relative to a fixed origin O .]

At midnight, a ship S is at the point with position vector $(19\mathbf{i} + 22\mathbf{j})\text{ km}$

The ship travels with constant velocity $(12\mathbf{i} - 16\mathbf{j})\text{ km h}^{-1}$

- (a) Find the speed of S .

(2)

At time t hours after midnight, the position vector of S is $\mathbf{s}\text{ km}$.

- (b) Find an expression for $\mathbf{s}$ in terms of $\mathbf{i}$, $\mathbf{j}$ and t .

(2)

A lighthouse stands on a small rocky island. The lighthouse is modelled as being at the point with position vector $(26\mathbf{i} + 15\mathbf{j})\text{ km}$.

It is not safe for ships to be within 1.3 km of the lighthouse.

- (c) (i) Find the value of t when S is closest to the lighthouse.

- (ii) Hence determine whether it is safe for S to continue its course.

(7)

Q38 PAST PAPER Pearson IAL WME01 • June 2024 • Q7

M1 • TOPIC 2 13 marks

7. [In this question, the horizontal unit vectors $\mathbf{i}$ and $\mathbf{j}$ are directed due east and due north respectively and position vectors are given relative to a fixed origin O .]

Two speedboats, A and B , are each moving with constant velocity.

- the velocity of A is 40 km h^{-1} due east
- the velocity of B is 20 km h^{-1} on a bearing of angle α ($0^\circ < \alpha < 90^\circ$), where $\tan \alpha = \frac{4}{3}$

The boats are modelled as particles.

- (a) Find, in terms of $\mathbf{i}$ and $\mathbf{j}$, the velocity of B in km h^{-1} (2)

At noon

- the position vector of A is $20\mathbf{j}$ km
- the position vector of B is $(10\mathbf{i} + 5\mathbf{j})$ km

At time t hours after noon

- the position vector of A is $\mathbf{r}$ km, where $\mathbf{r} = 20\mathbf{j} + 40t\mathbf{i}$
- the position vector of B is $\mathbf{s}$ km

- (b) Find an expression for $\mathbf{s}$ in terms of t , $\mathbf{i}$ and $\mathbf{j}$. (2)

- (c) Show that at time t hours after noon,

$$\overrightarrow{AB} = [(10 - 24t)\mathbf{i} + (12t - 15)\mathbf{j}] \text{ km} \quad (2)$$

- (d) Show that the boats will never collide. (3)

- (e) Find the distance between the boats when the bearing of B from A is 225° (4)

Q39**PAST PAPER**

Pearson IAL WME01 · January 2026 · Q6

M1 · TOPIC 2 16 marks

6. [In this question $\mathbf{i}$ and $\mathbf{j}$ are horizontal unit vectors directed due east and due north respectively and position vectors are given relative to a fixed origin O .]

At 9 am, a ship S leaves a harbour and moves with constant velocity $(10\mathbf{i} + 15\mathbf{j}) \text{ km h}^{-1}$

- (a) Find the exact speed of S

(2)

The harbour is at the origin O .

At 9 am, a boat B leaves the point with position vector $(50\mathbf{i} + 10\mathbf{j}) \text{ km}$ and moves with constant velocity $(-8\mathbf{i} + 10\mathbf{j}) \text{ km h}^{-1}$

- (b) Find the direction in which B is moving, giving your answer as a bearing to the nearest degree.

(3)

- (c) Show that at time t hours after 9 am,

$$\overrightarrow{SB} = [(50 - 18t)\mathbf{i} + (10 - 5t)\mathbf{j}] \text{ km}$$

(4)

- (d) Show that B and S do not collide.

(3)

When B is north-east of S , the distance between S and B is $d \text{ km}$.

- (e) Find the value of d .

(4)

ALL The Whole Topic on One Page

Tear this page out before the exam.

- | | | |
|-----------|---------------------------|---|
| 01 | Size and direction | Sketch; Pythagoras; $\tan$; bearing clockwise from north. |
| 02 | Forces | Sum = 0 or ma ; parallel $\rightarrow$ ratio of parts. |
| 03 | Constant velocity | $\mathbf{r} = \mathbf{r}_0 + \mathbf{v}t$; due east $\rightarrow$ $\mathbf{j}$ part 0. |
| 04 | Two objects | $\mathbf{b} - \mathbf{a}$; collide: both parts 0; closest: complete the square. |

BOTTOM LINE Sketch every vector $\cdot \mathbf{i}$ with $\mathbf{i}$, $\mathbf{j}$ with $\mathbf{j}$ $\cdot$ closest approach = minimum of $|\vec{AB}|^2$.

Answers

Working with Vectors • for the teacher. Every answer checked with a computer algebra system.

IDEA 01 • MAGNITUDE, DIRECTION AND BEARINGS

- Q1** Dr Eslam • Magnitude and bearing
5 N; bearing $360^\circ - 53.1^\circ = 306.9^\circ$
- Q2** Dr Eslam • From a bearing to $\mathbf{i}, \mathbf{j}$
 $10 \sin 60^\circ \mathbf{i} + 10 \cos 60^\circ \mathbf{j} \approx 8.66\mathbf{i} + 5\mathbf{j}$
- Q3** Dr Eslam • Angle between two vectors
directions $\tan^{-1} \frac{4}{3} = 53.1^\circ$ and 45° : angle 8.1°
- Q4** Pearson IAL WME01 • October 2019 • Q7
(i) $\sqrt{52} \approx 7.21$ N (ii) 074°
- Q5** Pearson IAL WME01 • January 2020 • Q6
(a) $\sqrt{101}$ N (b) 39.3° (c) $a = 6.5$
- Q6** Pearson IAL WME01 • January 2021 • Q5
(i) $\sqrt{244} \approx 15.6$ N (ii) 206°
- Q7** Pearson IAL WME01 • January 2024 • Q4
(i) $X = 18$ (ii) 128°
- Q8** Pearson IAL WME01 • June 2024 • Q2
 $|\mathbf{F}| = 2\sqrt{61} \approx 15.6$ N
- Q9** Pearson IAL WME01 • June 2026 • Q6
(a) $\sqrt{61}$ N (b) 103° (c) $\mathbf{F}_1 = 3\mathbf{i} - 6\mathbf{j}$, $\mathbf{F}_2 = -8\mathbf{i} + 12\mathbf{j}$

IDEA 02 • FORCES AS VECTORS: EQUILIBRIUM AND $\mathbf{F} = m\mathbf{a}$

- Q10** Dr Eslam • Three forces in equilibrium
 $p = 3, q = -2$
- Q11** Dr Eslam • Acceleration from forces
 $\mathbf{a} = 2\mathbf{i} - 2\mathbf{j}; 2\sqrt{2} \approx 2.83 \text{ m s}^{-2}$
- Q12** Dr Eslam • Parallel to a vector
 $2(p + 3) = p + 2 \Rightarrow p = -4$
- Q13** Pearson IAL WME01 • June 2021 • Q3
 (a) 33.7° (b) $\lambda = 3$
- Q14** Pearson IAL WME01 • January 2022 • Q6
 (a) $p = -6, q = 4$ (b) 135° (c) $T = 2.5$
- Q15** Pearson IAL WME01 • October 2022 • Q6
 (a) $2p + q - 4 = 0$ (b) $16\sqrt{5} \approx 35.8 \text{ m s}^{-1}$
- Q16** Pearson IAL WME01 • January 2023 • Q3
 (a) $3\sqrt{5} \approx 6.7 \text{ m s}^{-1}$ (b) 27° (c) $T = 3.2$
- Q17** Pearson IAL WME01 • June 2023 • Q2
 (a) $\mathbf{F}_3 = (14 - 3c)\mathbf{i} + (-7 - 4c)\mathbf{j}$ (b) 118° (c) $m = 2$
- Q18** Pearson IAL WME01 • January 2025 • Q1
 (a) $\frac{12\sqrt{13}}{5} \approx 8.7 \text{ m s}^{-2}$ (b) $p = -30$
- Q19** Pearson IAL WME01 • October 2025 • Q3
 (a) $\mathbf{i} + 2.5\mathbf{j}$ (b) $\frac{\sqrt{193}}{2} \approx 6.95 \text{ m s}^{-1}$ (c) $b - c = 3$
 (d) $8\mathbf{i} + 5\mathbf{j}$ or $-12\mathbf{i} - 15\mathbf{j}$
- Q20** Pearson IAL WME01 • January 2026 • Q3
 (a) $2p - q = 19$ (b) $24\mathbf{i} - 30\mathbf{j} \text{ m s}^{-2}$

IDEA 03 • CONSTANT VELOCITY: $\mathbf{r} = \mathbf{r}_0 + \mathbf{v}t$

- Q21** Dr Eslam • Where and when
 $\mathbf{r} = (2 + 3t)\mathbf{i} + (5 - t)\mathbf{j}$; due east when $t = 5$: $17\mathbf{i}$
- Q22** Dr Eslam • Velocity from two positions
 $\mathbf{v} = 3\mathbf{i} - 4\mathbf{j}$; 5 m s^{-1}
- Q23** Pearson IAL WME01 • October 2021 • Q4
 (a) 162° (b) $t = \frac{1}{2}$ or $\frac{3}{2}$
- Q24** Pearson IAL WME01 • October 2021 • Q8
 (a) $10\mathbf{i} - 5\mathbf{j}$ (b) $(-2 + 10t)\mathbf{i} + (9 - 5t)\mathbf{j}$ (c) $19\mathbf{i}$ (d) $19\mathbf{i} - 1.5\mathbf{j}$
- Q25** Pearson IAL WME01 • October 2024 • Q3
 (a) $\mathbf{r}_A = (25 + 15t)\mathbf{i} + (10 + 12t)\mathbf{j}$ (b) $v = \frac{5\sqrt{41}}{6} \approx 5.34$ (c) A is at P at $t = \frac{8}{3}$
- Q26** Pearson IAL WME01 • January 2025 • Q5
 (a) $1.5\mathbf{i} - \mathbf{j} \text{ m s}^{-1}$ (b) $2\sqrt{13} \approx 7.2 \text{ m}$
- Q27** Pearson IAL WME01 • June 2025 • Q2
 (a) $5\mathbf{i} - 4\mathbf{j}$ (b) $(5t - 15)\mathbf{i} + (24 - 4t)\mathbf{j}$ (c) 203°
- Q28** Pearson IAL WME01 • June 2026 • Q7
 (a) $\mathbf{r} = (14 - 2t)\mathbf{i} + (2 + t)\mathbf{j}$ (b) 3.6 hours (c) $t = 16$ (d) $\frac{18\sqrt{5}}{5} \text{ km}$

IDEA 04 • TWO MOVING OBJECTS: MEET, CLOSEST, BEARING

- Q29** Dr Eslam • Collide or not? Closest? $\vec{AB} = (10 - 2t)\mathbf{i} + 2t\mathbf{j}$; no collision; $|AB|^2 = 8t^2 - 40t + 100$, least at $t = 2.5$: $\sqrt{50} \approx 7.07$ km
- Q30** Pearson IAL WME01 • October 2019 • Q8 (a) $20\mathbf{i} - 20\sqrt{3}\mathbf{j}$ (b) $\mathbf{r} = (60 - 20t)\mathbf{i}$, $\mathbf{s} = 20t\mathbf{i} - 20\sqrt{3}t\mathbf{j}$ (c) 13:43
- Q31** Pearson IAL WME01 • January 2021 • Q6 (a) $(11 + 3t)\mathbf{i} + (11 - t)\mathbf{j}$ (b) $t = 5.5$ (c) shown (d) $\sqrt{0.5} \approx 0.707$ m
- Q32** Pearson IAL WME01 • January 2022 • Q8 (a) $(13 + 3t)\mathbf{i} + (5 - 10t)\mathbf{j}$ (b) shown (c) $2\sqrt{5} \approx 4.5$ km (d) 297°
- Q33** Pearson IAL WME01 • June 2022 • Q8 (a) 135° (b) shown (c) 270° (d) $40\sqrt{2}$ m (e) $t = 0$ or $\frac{320}{17}$
- Q34** Pearson IAL WME01 • October 2022 • Q8 (a) $3\sqrt{17}$ km h⁻¹ (b) $p = -9$, $q = 3$ (c) 323°
- Q35** Pearson IAL WME01 • June 2023 • Q8 (a) $(t - 2)\mathbf{i} + (2t + 1)\mathbf{j}$ (b) $(25 - t)\mathbf{i} + (25 - t)\mathbf{j}$ (c) shown (d) $t = \frac{126}{13} \approx 9.7$ s
- Q36** Pearson IAL WME01 • October 2023 • Q6 (a) $10\mathbf{i} - 4\mathbf{j}$ (b) $(15 + 10t)\mathbf{i} + (36 - 4t)\mathbf{j}$ (c)(i) both at $30\mathbf{i} + 30\mathbf{j}$ when $t = 1.5$ (ii) $30\mathbf{i} + 30\mathbf{j}$ (d) $2\sqrt{101}$ km
- Q37** Pearson IAL WME01 • January 2024 • Q7 (a) 20 km h⁻¹ (b) $(19 + 12t)\mathbf{i} + (22 - 16t)\mathbf{j}$ (c)(i) $t = 0.49$ (ii) least distance 1.4 km > 1.3: safe
- Q38** Pearson IAL WME01 • June 2024 • Q7 (a) $16\mathbf{i} + 12\mathbf{j}$ (b) $(10 + 16t)\mathbf{i} + (5 + 12t)\mathbf{j}$ (c) shown (d) never both zero (e) $\frac{20\sqrt{2}}{3} \approx 9.43$ km
- Q39** Pearson IAL WME01 • January 2026 • Q6 (a) $5\sqrt{13}$ km h⁻¹ (b) 321° (c) shown (d) do not collide (e) $d = \frac{70\sqrt{2}}{13} \approx 7.6$ km